WORKED SOLUTION

STEP III 2011, Question 8

Pure • Complex Loci

STEP III 2011, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q8

The key idea

Multiply top and bottom by the conjugate of the denominator to get u and v; after that, each part is a matter of substituting the given line and spotting a familiar curve. The half-angle substitution turns $\frac{2x}{1+x^2}$ and $\frac{x^2-1}{x^2+1}$ into a sine and a cosine, which is exactly why the question suggests it. Throughout, keep an eye on *which* points of the curve are actually reached: the endpoints and missing points are half the answer.

Solution

Write $z = x + \mathrm{i}y$. Then $1 + \mathrm{i}z = (1 - y) + \mathrm{i}x$ and $\mathrm{i} + z = x + \mathrm{i}(1 + y)$, so, multiplying numerator and denominator by $x - \mathrm{i}(1 + y)$,

$$\begin{aligned} w &= \frac{((1-y)+\mathrm{i}x)(x-\mathrm{i}(1+y))}{x^2+(1+y)^2} \\ &= \frac{(1-y)x + x(1+y) + \mathrm{i}(x^2-(1-y)(1+y))}{x^2+(1+y)^2}. \end{aligned}$$

Hence

$$u = \frac{2x}{x^2+(1+y)^2}, \qquad v = \frac{x^2+y^2-1}{x^2+(1+y)^2},$$

valid for $z \neq -\mathrm{i}$ (where w is undefined).

It also helps to have the inverse map. From $w(\mathrm{i} + z) = 1 + \mathrm{i}z$ we get $z(w - \mathrm{i}) = 1 - \mathrm{i}w$, so

$$z = \frac{1 - \mathrm{i}w}{w - \mathrm{i}}, \qquad w \neq \mathrm{i}.$$

So every $w \neq \mathrm{i}$ comes from exactly one z, and $w = \mathrm{i}$ is never attained. That is the omitted point we shall keep meeting.

Part (i)

With $y = 0$,

$$u = \frac{2x}{1+x^2}, \qquad v = \frac{x^2-1}{x^2+1}.$$

Put $x = \tan(\theta/2)$. As x runs over all real numbers, θ runs over $-\pi < \theta < \pi$, and each x gives exactly one such θ. By the half-angle formulae,

$$u = \frac{2\tan(\theta/2)}{1+\tan^2(\theta/2)} = \sin\theta, \qquad v = -\frac{1-\tan^2(\theta/2)}{1+\tan^2(\theta/2)} = -\cos\theta.$$

So $u^2 + v^2 = 1$. Moreover $(\sin\theta, -\cos\theta)$ for $-\pi < \theta < \pi$ traces the whole unit circle except the point given by $\theta = \pm\pi$, namely $(u, v) = (0, 1)$. So the locus of w is the circle $u^2 + v^2 = 1$ with the single point $w = \mathrm{i}$ omitted, as required. (The omitted point is the limit of w as $x \to \pm\infty$.)

Part (ii)

Now $-1 < x < 1$, which corresponds to $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. On this range $v = -\cos\theta < 0$, and $u = \sin\theta$ increases from -1 to 1 (not inclusive). So the locus of w is

$$u^2 + v^2 = 1, \quad v < 0,$$

the lower half of the unit circle, *not* including its endpoints $w = \pm 1$ (which come from $x = \pm 1$).

Part (iii)

With $x = 0$ we get $u = 0$ and

$$v = \frac{y^2-1}{(1+y)^2} = \frac{(y-1)(y+1)}{(y+1)^2} = \frac{y-1}{y+1} = 1 - \frac{2}{y+1},$$

where cancelling $y + 1$ is fine because $y > -1$.

? **Ask yourself** Is every negative value of v reached, or only some of them?

As y increases from -1 to 1, $y + 1$ increases through $(0, 2)$, so $\frac{2}{y+1}$ decreases through $(1, \infty)$, and v increases continuously and strictly through $(-\infty, 0)$. As $y \to -1^+$, $v \to -\infty$, and as $y \to 1^-$, $v \to 0^-$. So every negative v is reached exactly once. The locus of w is the negative imaginary axis

$$u = 0, \quad v < 0,$$

with the origin excluded (it comes from $y = 1$).

Part (iv)

With $y = 1$,

$$u = \frac{2x}{x^2+4}, \qquad v = \frac{x^2}{x^2+4}.$$

Then

$$u^2+v^2=\frac{4x^2+x^4}{(x^2+4)^2}=\frac{x^2(x^2+4)}{(x^2+4)^2}=\frac{x^2}{x^2+4}=v,$$

so w lies on $u^2+v^2=v$, that is,

$$u^2+(v-\tfrac12)^2=\tfrac14,$$

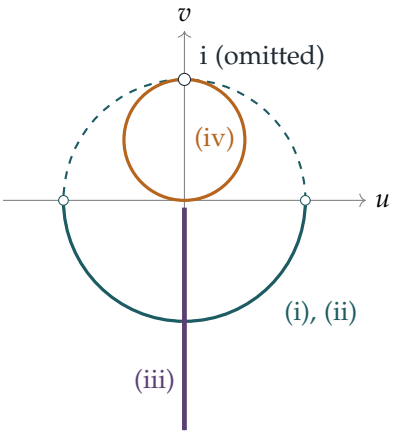
the circle with centre $\frac12\mathrm{i}$ and radius $\frac12$.

Ask yourself Which points of this circle are actually reached?

Since $v=1-\frac{4}{x^2+4}$, we have $0\le v<1$, so the top point $w=\mathrm{i}$ is never reached. Every other point is: setting $x=2\tan(\phi/2)$ with $-\pi<\phi<\pi$ gives

$$u=\frac{\tan(\phi/2)}{1+\tan^2(\phi/2)}=\tfrac12\sin\phi,\qquad v=\frac{\tan^2(\phi/2)}{1+\tan^2(\phi/2)}=\tfrac12(1-\cos\phi),$$

which runs round the whole circle except $\phi=\pm\pi$, the point $(0,1)$. (Alternatively, the inverse map shows that each $w\neq\mathrm{i}$ comes from exactly one z.) So the locus is the circle $u^2+v^2=v$ with the point $w=\mathrm{i}$ omitted. It touches the real axis at $w=0$, the image of $z=\mathrm{i}$.



The dashed and solid teal circle is the image of the real axis, part (i); the solid lower half is part (ii). The plum ray is part (iii) and the amber circle is part (iv).

Checks

- $z=0$ gives $w=1/\mathrm{i}=-\mathrm{i}$: on the lower semicircle (ii) and on the ray (iii), and indeed $z=0$ lies on both segments.
- $z=1+\mathrm{i}$ gives $w=\frac{\mathrm{i}}{1+2\mathrm{i}}=0.4+0.2\mathrm{i}$, and $0.4^2+0.2^2=0.2=v$, so it lies on the circle in (iv).

- Large real z: $w(10^8) \approx 2 \times 10^{-8} + \mathrm{i}$, approaching the omitted point i.
- We also checked numerically, for thousands of random points, both the formulae for u, v and that every point of each claimed locus has a preimage on the right line or segment.

Common slips

- Giving the circle or semicircle but not saying which points are omitted or excluded: here these are $w = \mathrm{i}$ in (i) and (iv), and the endpoints ± 1 in (ii).
- Claiming the locus in (iii) is the whole imaginary axis: $v = \frac{y-1}{y+1}$ is negative for every $-1 < y < 1$.

Beyond STEP

The map is a Möbius transformation; in fact $w = \mathrm{i}\,\dfrac{z-\mathrm{i}}{z+\mathrm{i}}$, so $|w| = |z-\mathrm{i}|/|z+\mathrm{i}|$, which equals 1 exactly when z is equidistant from $\pm\mathrm{i}$, i.e. real. Möbius maps send lines and circles to lines and circles (a line being a "circle through ∞"), and the omitted point i is the image of ∞. This map (a Cayley transform) sends the upper half-plane onto the unit disc, a key tool in complex analysis and hyperbolic geometry. Needham's *Visual Complex Analysis* has a beautiful, picture-led treatment of Möbius transformations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.